devices, reducing wastage from the leftover substrate. Hence, the reliability of a component extracted out of the substrate increases.

Insights

Ruggedised components: New SiC power devices with trench structure

Ruggedised PCB: Reduced thickness and impregnation with new material

Ruggedised mobile devices: MEMS sensors giving compact and rugged ADAS solution

Ruggedised networking: Wireless networking protocols

Sensing solutions and mobile solutions

Smart applications, including mobile applications and their components, and networking applications and their hardware, have been customised at the embedded level.

MEMS sensors. Kamat says, "MEMS sensors and nanotechnology are next-level technologies in the industry."

MEMS sensors are being used widely in mobile divisions, smart automation systems, imaging systems and microphone applications. Advanced driver-assistance system (ADAS) in automobiles accounts for the largest market. It combines accelerator, gyroscope and speedometer in one system to give an error-free, compact solution, saving the large area which each of these functions used to cover through the driving circuitry and wire mesh.

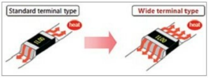

Fig. 3: Wide terminal configuration for high joint reliability

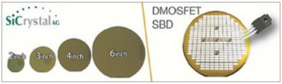

Fig. 4: The increased substrate size (15cm) helps in churning more number of devices

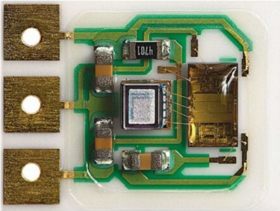

Fig. 5: MEMS pressure sensors (Image courtesy: http://4al7et3fds5z3zj5fm3mbpds.wpengine.netdna-cdn.com)

MEMS sensors make the hardware system of automobiles rugged as well as reliable. Kamat cites the example of sensor-based wheels, wherein MEMS sensors inform the driver about a fault in advance, preventing accidents. Automobile giant Tesla has converted its electric car system to sensor-based. Thus, while no use of fuel reduces the chances of pollution, the sensor-based technology ensures fast and reliable functioning.

Talking about wearables, SOLI sensors are taking the market by storm. These sensors easily recognise 3D gestures expressed with the help of fingers and hands. Low-bandwidth application and small size of SOLI sensors make these a reliable component for wearable devices.

Networking protocols reducing the use of wired connections. Smart city, a project picking up pace worldwide, relies on the Internet of Things (IoT) wireless network system. Wireless system is believed to replace the LoRaWAN and NBIoT system. Thus, the use of sensor technology and wireless network has made it possible for the future technology to gear up and increase the bandwidth.

Low latency, high bandwidth, long life-cycle and low power consumption make WiSUN a reliable network protocol that is easier for the enterprises to accept over time due to its fail-safe communication.

Lithium-polymer battery replacing Li-ion battery. Since electric cars have become the next-generation vehicle, designers are focused on increasing the longevity and ruggedness of their batteries.

Compact size and low weight are making lithium polymer batteries the future trend. Due to their large current-carrying capacity, these batteries can be charged several times without delay. This makes them a reliable input component of mobile phones.

Kamat says, "Lithium polymer batteries supply a large current and are compact in size, which makes these affordable for smartphones."

Testing the hardware ruggedness

Hardware ruggedness testing includes testing of the hardware against extreme temperatures, shock and vibrations, and other environmental conditions. When a product is designed, it is passed through various harsh test conditions in order to confirm its quality and reliability.

After design phase, the product undergoes test and analysis, where it is checked thoroughly for its electrical, mechanical and physical properties.

Ruggedness is a critical feature for wearables, sensors, power supply devices and components. If manufacturers are careless during design and development phase, the end-product would neither be reliable nor be able to create its value in the market. EFY